Mingzhe Li

Robotics Systems / Embodied AI / 3D Vision & Robot Learning
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Summary

Incoming M.Sc. student in Robotics and Autonomous Systems (RAS) at the University of Macau (Fall 2026). Researcher–engineer in robotics systems, embodied AI, and 3D vision, with end-to-end experience from hardware prototyping and perception/navigation to teleoperation data collection, VLA policy fine-tuning, and long-horizon task agents. Currently at INFIFORCE Shenzhen Research Institute on home-service robots, and a core research intern in Prof. Jiayu Chen's Agentic Intelligence Lab at the University of Hong Kong. Contributor to an ICML 2025 paper. During undergraduate studies, founded BNBU's first student robotics lab, Navigator Robotics Lab (grew to ∼35 members), and led the team to 1st Place in the RoboMaster Regional Engineer Challenge (Liaoning).

Research Interests

Embodied AI & Robot Learning: vision–language–action models, imitation learning, manipulation, long-horizon planning, world models, task agents, cross-embodiment adaptation
3D Vision & Spatial Perception: 3D Gaussian Splatting, NeRF, SLAM, dynamic reconstruction, semantic mapping, geometry-consistent and interactive scene representations
Robotics Systems & Scene Agents: real-robot integration, teleoperation & data pipelines, FSM/tool-calling orchestration, multimodal HRI
Neuro-symbolic & Generative Engineering: parametric CAD generation, structured code generation, symbolic constraint solving, domain LMs for engineering design

Education

University of Macau — Macau, China
Incoming M.Sc. in Robotics and Autonomous Systems — *Aug 2026 – Jun 2028*

Beijing Normal–Hong Kong Baptist University (BNBU / UIC) — Zhuhai, China
B.Sc. (Honours) in Data Science — *Aug 2022 – Jun 2026*

Research & Work Experience

INFIFORCE Shenzhen Research Institute — Shenzhen, Chin
Research Assistant / Embodied AI Systems Engineering Intern (Agentic Intelligence Lab, Prof. Jiayu Chen, HKU)Mar 2026 – Sep 2026 (expecte

- Work on home-service robots and VLA pipelines: deployment → real data collection → policy training → task execution feedback.
- Build robot prototypes end-to-end (mechanics, PCB, wiring, drive calibration, CAN, integration); adapt AgileX Piper, Nero, OpenArm, Franka, Galaxea R1, etc.
- Integrate open-source stacks for SLAM/navigation, vision, calibration, VR teleop, and training; extend from single-/dual-arm to whole-body systems.
- Fine-tune ACT and OpenPI $\pi_{0.5}$ with LeRobot for manipulation learning and cross-embodiment adaptation.
- Own long-horizon task-agent planning/execution; lead reviews for design, drawings, PCBs, and bring-up in a small team.

Selected Robotics Projects

Furniture-Scale Mobile 3D Printing Platform | *Independent / In Progress* — Jul 2026 – Jan 2027 (expected)

- **Gap:** Consumer printers (e.g., Bambu Lab, Creality) are typically within ∼30 cm build size; larger scales jump to costly industrial/specialty-material AM or building-scale cement printers. The ∼0.3–3 m furniture/indoor scale is an awkward gap in size, materials, nozzles, and cost.
- **Goal:** Build a mobile indoor printing platform and complete a 100-day challenge: develop the system and achieve stable furniture-scale printing (a Lego-style systems challenge).
- **System:** Planned Gaoqing Tech arm on a mobile base (AgileX / Slamtec or self-built), with fused indoor localization (UWB, visual markers, LiDAR, wheel odometry) for long-horizon continuous printing.
- **Pipeline:** voice/agent intent → STL generation → slicing & nozzle path planning → G-code execution, including kinematics, collision avoidance, extrusion control, and full-stack mech/URDF/sim/electronics integration.

- **Status:** Just started (concept & platform selection). Submodules are largely available; the hard part is system coupling and long-duration reliability.
- **Tech:** mobile manipulator, localization fusion, path planning, slicing/G-code, motion control, ROS, SolidWorks, 3D-printing system integration.

PhotoMate: Mobile Dual-Arm Embodied Photography Service | *Lead / System Design & Integration* Jul 8–11, 2026

- Designed a long-horizon task agent (S0–S6) for on-site photography: proactive invite → viewpoint guidance → multimodal interaction → capture QA → QR delivery.
- Delivered a runnable software stack: FastAPI session/FSM, Qwen-Omni multimodal interaction with function calling, React frontend, capture/upload/QR pipeline (open-sourced).
- Abstracted navigation, perception, arm/hand, camera, and interaction as swappable capability interfaces; explicit FSM owns timeouts, retries, and resource release.
- Real-robot closed loop incomplete because sponsor platform did not expose required navigation/manipulation APIs; validated product flow, architecture, and software chain—not a full field demo.
- **Tech:** mobile dual-arm robot, task agent/FSM, FastAPI, Qwen-Omni, React, OpenCV, SolidWorks, 3D printing, system integration.
- Page: photomate.html; Code: PhotoMate-Moonbot-Hackthon.

Robotics & Engineering Experience

Navigator Robotics Lab, BNBU Zhuhai, China
Founder / Technical Lead *Apr 2023 – Aug 2025*

- Founded the university's first student robotics lab; grew to 35 members with ∼¥200k funding.
- Led platform design and full-stack delivery across mechanics, embedded, vision, algorithms, and competitions.
- Integrated industrial cameras, LiDAR, Jetson, ROS2, and real-time control for combat, engineer, and wheeled-arm robots.
- 1st Place, RoboMaster University League Engineer Challenge (Liaoning Regional).

Other Research Experience

IsoGS-SLAM: Geometry-Consistent Real-Time 3D Reconstruction via Surface-Aligned Gaussians | *Lead*Dec 2025 – Apr 2026

- Study geometric consistency issues of 3D Gaussian Splatting for embodied interaction (surface geometry, normal continuity, physical interaction).
- Introduce implicit geometric constraints (e.g., Metaballs) into explicit Gaussians via variational optimization, analytic normal alignment, and curvature-guided densification.
- Aim for a unified scene representation supporting real-time rendering, dense geometry, collision checking, and manipulation planning.
- **Tech:** PyTorch, CUDA, differentiable Gaussian rasterization, 3DGS, SLAM, geometric optimization.

Domain Language Model for Parametric CAD Parts & Assemblies | *Independent Research* Nov 2025 – Mar 2026

- Built ∼35,000 synthetic parametric CAD samples (∼20k parts + ∼15k assemblies) for code-generation training.
- LoRA-tuned DeepSeek Coder 6.7B: 95.5% syntactic correctness, 88.2% execution success, ∼86–92% constraint satisfaction across tasks.
- Explored neuro-symbolic coupling of LMs with symbolic solvers for generation, assembly relations, and engineering rules.
- **Tech:** PyTorch, LoRA, DeepSeek Coder, CadQuery, parametric modeling, symbolic constraints.

Robust SLAM & Object Tracking in Dynamic Environments | *Undergraduate Thesis* May 2025 – Nov 2025

- Combined YOLO semantic segmentation with ORB-SLAM3 to suppress dynamic objects for pose estimation and mapping.
- Studied real-time dynamic scene reconstruction with 3DGS, jointly representing ego-motion and moving-object trajectories.
- Designed evaluation metrics for ego-motion, tracking, and reconstruction quality. **Advisor:** Prof. Zhe Xuanyuan.

Publications

Demonstration Selection for In-Context Learning via Reinforcement Learning **ICML 2025**

- Proposed RDES, casting in-context demonstration selection for LLMs as a sequential decision process via reinforcement learning.
- Contributed to core algorithm development, experiments, analysis, and research documentation.

Other Internships

WISPRO / Xiaomei Intellectual Property Group Shenzhen, China
LLM Engineering Intern *Jun 2025 – Aug 2025*

- Built LangChain multi-agent workflows for patent document processing, automation, retrieval, and collaboration.
- Supported GPU cluster management, training, deployment, and stability validation.

EpicHust Technology (Wuhan) Co., Ltd. Wuhan, China
LLM Engineering Intern *Jan 2025 – Feb 2025*

- Fine-tuned and evaluated open-source LMs for industrial protocols and smart-factory use cases; SFT on Qwen2.5/DeepSeek; RAG and knowledge-base design.

Skills

Embodied AI & VLA: LeRobot, ACT fine-tuning, OpenPI $\pi_{0.5}$, robot data collection, VR teleop, imitation learning, task planning, long-horizon agents, multi-agent systems
Robotics Systems: ROS1/ROS2, Gazebo, SLAM, localization/mapping/navigation, CAN, motor drives, arm control, end-effectors, camera calibration, sensor fusion
3D Vision & Neural Rendering: 3DGS, NeRF, ORB-SLAM3, FAST-LIVO2, dynamic reconstruction, geometric optimization, semantic/instance 3D perception
ML & LLMs: PyTorch, Transformers, RL, LoRA, SFT, RAG, LangChain, Qwen-Omni, in-context learning, code generation, neuro-symbolic systems
Programming & Tools: Python, C++, CUDA, Linux, Git, Docker, FastAPI, Flask, React, SQL, LaTeX
Hardware & Prototyping: SolidWorks, OnShape, parametric CAD, 3D printing, PCB, assembly, wiring, industrial/depth cameras, LiDAR, Jetson
Platforms: AgileX Piper, Nero, OpenArm, Franka, Galaxea R1, wheeled dual-arm platforms, Jetson Orin
Languages: Mandarin (native); English for coursework, papers, technical writing, and professional communication; Cantonese for daily and basic professional use

Honors

- 1st Place, RoboMaster University League Engineer Challenge (Liaoning Regional), 2024.
- 2nd Prize, Greater Bay Area Cup Financial Mathematical Modeling Contest (AI for Science), 2024.
- MCM/ICM Honorable Mention, 2024.
- 1st Prize, Huazhong Cup Mathematical Modeling Challenge, 2023.